DAY 2 – Hands On

Pre-Requisites: Before starting with problem solving, please make sure that you have created a database and restored data (refer **Sample Database** create objects.txt and load data.txt)

Level-1: Problem 1 - Basic Customer Order Report

Scenario:

The store manager wants a simple report showing customer orders along with their order dates and status. This report will help track pending and completed orders.

📌 Requirements

1. Retrieve customer first name, last name, order_id, order_date, and order_status.
2. Display only orders with status Pending (1) or Completed (4).
3. Sort the results by order_date in descending order.

🛠 Technical Constraints

- Use SELECT statement.
- Use WHERE clause with logical operators (AND/OR).
- Use ORDER BY clause.
- Use INNER JOIN between customers and orders tables.

Expectations:

Students should write a correct query using joins and filters, and properly order the result set.

🎯 Learning Outcome

Understand basic SELECT queries, filtering using WHERE conditions, logical operators, and sorting using ORDER BY clause with INNER JOIN.

Level-1: Problem 2 - Product Price Listing by Category

Scenario:

The sales team wants a product listing categorized by product category along with brand details to understand product distribution.

📌 Requirements

1. Display product_name, brand_name, category_name, model_year, and list_price.
2. Filter products with list_price greater than 500.
3. Sort results by list_price in ascending order.

🛠 Technical Constraints

- Use SELECT statement.
- Use WHERE clause.
- Use ORDER BY clause.
- Use INNER JOIN between products, brands, and categories tables.

Expectations:

Students should correctly join multiple tables and apply filtering and sorting logic.

🎯 Learning Outcome

Learn multi-table joins, filtering numeric conditions, and sorting query results.

Level-2: Problem 1 - Store Wise Sales Summary

Scenario:

The regional manager wants to analyze total sales generated by each store.

📌 Requirements

1. Display store_name and total sales amount.
2. Calculate total sales using (quantity * list_price * (1 - discount)).
3. Include only completed orders (order_status = 4).
4. Group results by store_name.
5. Sort total sales in descending order.

🛠 Technical Constraints

- Use INNER JOIN between stores, orders, and order_items tables.

- Use GROUP BY clause.

- Use aggregate functions such as SUM().

- Use WHERE clause for filtering.

- Use ORDER BY clause.

Expectations:

Students must write a grouped query with calculated columns and apply proper filtering.

🎯 Learning Outcome

Understand GROUP BY, aggregate functions, calculated fields, and filtering grouped data.

Level-2: Problem 2 - Product Stock and Sales Analysis
Scenario:

The inventory manager wants to compare stock availability and total quantity sold for each product.

📌 Requirements

1. Display product_name, store_name, available stock quantity, and total quantity sold.

2. Include products even if they have not been sold (use appropriate join).

3. Group results by product_name and store_name.

4. Sort results by product_name.

🛠 Technical Constraints

- Use LEFT JOIN between stocks and order_items.

- Use INNER JOIN with products and stores tables.

- Use GROUP BY clause.

- Use SUM() aggregate function.

- Use ORDER BY clause.

Expectations:

Students should correctly use different join types (INNER and LEFT), grouping, and aggregate functions.

🎯 Learning Outcome

Develop strong understanding of different JOIN types (INNER, LEFT), aggregation, grouping data, and analyzing relational datasets.